

4(a)	R if choice given and one is incorrect carbonic acid ; hydrogencarbonate (ions) ; A bicarbonate (ions)	2
4(b)	<i>any one from:</i> similar / same, active site ; have the same catalytic amino acids ; similar / same, tertiary structure ; A same quaternary structure	1

Question	Answer	Marks																		
4(c)	allow pre-RNA for primary transcript either exon or intron statement required for mp2 to 7, one mark each row any three from: <table><tr><th>exons</th><th>introns</th></tr><tr><td colspan="2">1 both, are (nucleotide sequences that are) transcribed / form part of the primary transcript ;</td></tr><tr><td>2 (DNA/RNA) <u>coding</u> (sequences) R coding genes I exons have the genetic code</td><td>(DNA/RNA) <u>non-coding</u> (sequences) R non-coding genes</td></tr><tr><td>3 (RNA exons) needed / involved / AW, in translation or for formation of, polypeptides / proteins / carbonic anhydrase</td><td>not involved in translation / AW</td></tr><tr><td colspan="2">if mps 2 and 3 not gained 'exons code for proteins' or ora 1 mark</td></tr><tr><td>4 not removed, from primary transcript / from RNA / during gene splicing / during RNA splicing</td><td>removed, from primary transcript / from RNA /during gene splicing / during RNA splicing</td></tr><tr><td>5 (join to) form mRNA / contain mRNA codons A remain in mRNA</td><td>not part of mRNA / in primary transcript only</td></tr><tr><td>6 not involved in regulating activity of genes / AW</td><td>may have regulatory roles A other suggestions for role</td></tr><tr><td>7 AVP e.g. (as part of mRNA) leave the nucleus / go to cytoplasm / go to ribosomes</td><td>AVP e.g. remain in nucleus</td></tr></table> if <u>all</u> are differences and <u>all</u> statements are reversed (i.e. introns for exons and vice versa) max 2 marks	exons	introns	1 both, are (nucleotide sequences that are) transcribed / form part of the primary transcript ;		2 (DNA/RNA) <u>coding</u> (sequences) R coding genes I exons have the genetic code	(DNA/RNA) <u>non-coding</u> (sequences) R non-coding genes	3 (RNA exons) needed / involved / AW, in translation or for formation of, polypeptides / proteins / carbonic anhydrase	not involved in translation / AW	if mps 2 and 3 not gained 'exons code for proteins' or ora 1 mark		4 not removed, from primary transcript / from RNA / during gene splicing / during RNA splicing	removed, from primary transcript / from RNA /during gene splicing / during RNA splicing	5 (join to) form mRNA / contain mRNA codons A remain in mRNA	not part of mRNA / in primary transcript only	6 not involved in regulating activity of genes / AW	may have regulatory roles A other suggestions for role	7 AVP e.g. (as part of mRNA) leave the nucleus / go to cytoplasm / go to ribosomes	AVP e.g. remain in nucleus	3
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4(d)	<i>allow Hb or hb or Hgb for haemoglobin</i> <i>two marks for points in left column or two marks for points in right column</i> <table><tr><td>cytoplasm / cytosol ;</td><td>(bound to) cell surface membrane ;</td></tr><tr><td>close to haemoglobin (for H⁺ uptake) R in / on / with, hb or <i>idea of</i> being dispersed within cell because haemoglobin is distributed throughout ;</td><td>close to, anion exchanger / transport protein, for (easy) exit of hydrogen carbonate (ions) ;</td></tr><tr><td>quick for hydrogen carbonate (ions) to be transported out of cell / AW or H⁺ (from dissociation) binds rapidly to haemoglobin or cytoplasm provides water for reaction to form carbonic acid ;</td><td>able to, hydrate / react with, carbon dioxide as it enters cell ;</td></tr></table> <i>if stated cytoplasm <u>and</u> cell surface membrane, one mark</i> <i>reason can be taken from either column if correctly linked, one mark</i>	cytoplasm / cytosol ;	(bound to) cell surface membrane ;	close to haemoglobin (for H ⁺ uptake) R in / on / with, hb or <i>idea of</i> being dispersed within cell because haemoglobin is distributed throughout ;	close to, anion exchanger / transport protein, for (easy) exit of hydrogen carbonate (ions) ;	quick for hydrogen carbonate (ions) to be transported out of cell / AW or H ⁺ (from dissociation) binds rapidly to haemoglobin or cytoplasm provides water for reaction to form carbonic acid ;	able to, hydrate / react with, carbon dioxide as it enters cell ;	2
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4(e)	exocytosis ;	1						
4(f)	<i>any three from:</i> 1 acetazolamide / inhibitor, binds / attaches / AW, to, allosteric site / site other than active site ; R alternative active site 2 (which) changes, shape / tertiary structure / 3-D structure, of active site ; A active site distorted I protein structure <i>allow ecf for mp3 and 4 if competitive described (to max 2)</i> 3 substrate(s) / CO₂ / HCO₃⁻, cannot, enter / bind to / fit in / AW, active site ; A active site no longer complementary to, substrate / AW A enzyme substrate / ES, complexes cannot form A ESCs cannot form I 'substrate / CO₂ / HCO₃⁻ / 'cannot bind to enzyme' <i>without a link to active site</i> 4 product, not formed / released ; A named product 5 <i>ref. to</i> acetazolamide / inhibitor, leaves, allosteric site / enzyme ; I reversibly binds	3						